# TOWARDS GLOBAL GEOMETRY PRESERVATION IN TEXT EMBEDDING DISTILLATION

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## ABSTRACT

A text embedding teacher encodes knowledge in the geometry it induces over an entire corpus, but exhaustively supervising all pairwise relations is infeasible. Standard relational knowledge distillation (KD) instead forms its supervision support from examples that happen to co-occur in a mini-batch, making relation exposure independent of teacher relevance. We show that preserving global teacher geometry depends on two factors: alignment, requiring the student to accurately reproduce the selected relations, and coverage, requiring those relations to capture most of the relationships the teacher considers important. This motivates a more effective estimator for corpus-level geometry preservation, using a compact set of teacher-relevant relations to capture the teacher's most important structure. To this end, we introduce **Global Geometry Preservation Knowledge Distillation (GGP-KD)**. GGP-KD builds a teacher-derived mutual kNN graph and uses diffusion to capture semantic structure at multiple scales. It then selects a compact set of informative relations and shares candidates across the mini-batch, allowing the student to learn both broader corpus structure and additional local relationships without extra encoding. Empirical results on nine standard sentence-embedding benchmarks across three teacher–student settings show that GGP-KD outperforms strong distillation baselines on both in-domain and out-of-domain benchmarks, with particularly strong gains on semantic textual similarity. GGP-KD requires no task labels or online teacher inference, and only the student is used at deployment.

## 1 INTRODUCTION

Text embedding models have become a fundamental component of modern NLP systems, supporting applications such as semantic search, retrieval, clustering, semantic textual similarity, and classification (Gao et al., 2023; Muennighoff et al., 2023). While larger embedding models can achieve strong performance, their computational and memory requirements may limit practical deployment, motivating the development of compact models that retain high-quality representations (Zhang et al., 2025). While larger text embedding models often achieve stronger performance on benchmark such as MTEB (Muennighoff et al., 2023), their increased parameter counts and representation dimensionality can incur substantial computational, memory, and deployment costs.

Knowledge distillation (KD) offers a practical way to reduce this deployment burden by transferring knowledge from a high-capacity teacher to a more compact student (Hinton et al., 2015). Recent embedding distillation frameworks have demonstrated strong empirical performance through direct teacher–student representation alignment (Vujanic & Rückstieß, 2026). For text embedding models, however, relational structure is also important, as many downstream applications operate directly on the similarities, distances, and relative ordering induced among embedded texts (Reimers & Gurevych, 2019; Muennighoff et al., 2023). Pointwise representation alignment does not explicitly constrain these inter-text relationships and may therefore leave important aspects of the teacher's embedding geometry uncontrolled. This has motivated relational and structure-preserving distillation methods that explicitly transfer teacher-induced similarities, distances, angles, and other geometric relationships (Tung & Mori, 2019; Park et al., 2019; Kim et al., 2023; Zhang et al., 2024; Dao et al., 2026).

At the corpus level, relational distillation would ideally account for the geometry induced among all training texts. Directly realizing such an objective, however, requires a quadratic number of pairwise relations. Relational KD is therefore commonly instantiated within mini-batches, where only relations among co-occurring texts are available for supervision (Tung & Mori, 2019; Park et al., 2019; Passalis et al., 2021). Prior work has identified this batch locality as a limitation: batch-wise objectives observe only a fraction of dataset-level relations, with relational coverage becoming increasingly sparse as the dataset grows (Qian et al., 2022; Jung et al., 2023). Full-kernel approximations and memory-augmented methods have consequently been explored to extend relational supervision beyond individual batches (Qian et al., 2022; Jung et al., 2023; Yang et al., 2022b). These approaches broaden relational supervision beyond individual batches, but do not address which corpus relations should be prioritized under a limited relational budget. We formulate this as a signal-selection problem: mini-batch KD lets batch co-occurrence determine relational support, whereas GGPKD lets teacher relevance select the relations and uses mini-batches only for computation.

We argue that this approximation introduces a under-emphasized bottleneck. Relational KD involves two distinct decisions: *which relations should be supervised* and *how those relations should be computed efficiently*. Mini-batch KD entangles the two. Although the relations themselves are defined by the teacher, their supervision support is determined by random batch co-occurrence. Consequently, an important teacher relation and an irrelevant corpus pair have the same probability of being exposed during training. Increasing optimization over such batches therefore does not directly address which parts of the teacher geometry are actually observed. This observation suggests a different principle: *the teacher should select the relations, while mini-batches should only compute them*. Based on this principle, we introduce GGPKD, a framework for teacher-selected global relational distillation. GGPKD first constructs a sparse mutual-nearest-neighbor graph from cached teacher embeddings, using teacher geometry to identify meaningful local relations. It then diffuses these local relations over the graph to construct multi-scale relational targets that capture structure beyond immediate neighbors. Finally, mass-aware candidate selection realizes these targets using only a small number of student comparisons, while cross-anchor supervision couples overlapping local views. The resulting procedure preserves teacher-selected corpus structure without requiring dense pairwise computation or online teacher inference. Our contributions are below:

- We formulate relational knowledge distillation as a *support-selection* problem and show that conventional mini-batch training selects relational supervision independently of teacher relevance.
- We propose GGPKD, which decouples teacher-driven relation selection from mini-batch computation through sparse teacher topology, multi-scale diffusion, and efficient candidate realization.
- We theoretically connect missed teacher supervision to global geometry distortion and empirically show that teacher-selected relation provides a more effective approximation to corpus-level relational transfer under a fixed relational budget.
- Experiment

## 2 Related Work

### 2.1 Text Embedding Distillation

Knowledge distillation transfers behavior from a large teacher to a compact student (Hinton et al., 2015). Language-model compression commonly matches output distributions, hidden states, or attention maps (Sanh et al., 2019; Jiao et al., 2020; Wang et al., 2020; Xu et al., 2024), whereas embedding-specific methods target quantities used by similarity-based applications. Sparse Logit Sampling shows that deterministic top-$K$ teacher logits induce truncation bias and uses randomized teacher-logit sampling to preserve the full teacher distribution in expectation (Anshumann et al., 2025). In that setting, however, the student still produces a full vocabulary vector; in corpus-level embedding KD, scoring an additional relation requires encoding another text, so support selection controls both approximation and encoder cost.

DistilCSE combines teacher supervision with contrastive learning (Gao et al., 2021; Xu et al., 2023); EmbedDistill transfers relative query–document geometry (Kim et al., 2023); and recent systems align embeddings, layers, tokens, or heterogeneous representation spaces (Lee et al., 2024; Zhang et al., 2024; Vujanic & Rueckstiess, 2025; Truong et al., 2025; Dao et al., 2026; Akram et al., 2026). Relational approaches match pairwise similarities, distances and angles, conditional affinities, graph edges, or topological summaries (Tung & Mori, 2019; Park et al., 2019; Passalis et al., 2021; Liu et al., 2019; Kim et al., 2024). Full Kernel Matrix Transfer uses Nystrom landmarks (Qian et al., 2022), and BINGO groups teacher-similar instances (Xu et al., 2022). Nevertheless, many relational objectives instantiate their relations from the current mini-batch; NRKD likewise mines an in-batch similarity matrix (Gou et al., 2025). This couples computational batching with geometric support selection.

### 2.2 Beyond Mini-batch Relational Supervision

Neighborhood graphs recover manifold structure in Laplacian Eigenmaps and Diffusion Maps (Belkin & Niyogi, 2003; Coifman & Lafon, 2006); heat-based constructions provide multi-scale summaries (Sun et al., 2009; Huguet et al., 2023). Geometric KD aligns heat kernels between teacher and student graphs (Yang et al., 2022a), while graph-to-MLP methods distill observed graph structure into graphless models (Zhang et al., 2022; Tian et al., 2025). RIPPLE instead constructs a graph from cached text embeddings to define supervision for a standard text encoder. It diffuses teacher relations before training and reuses pooled candidates as additional row centers; neither the graph nor the teacher is required at inference.

## 3 GGPKD: Global Geometry through Local Alignment

GGPKD distills an embedding teacher into a smaller student by matching the teacher's one-hop $k$NN transition rows. The method has only two method-specific hyperparameters: the graph width `graph_k` and the auxiliary-row weight `row_weight`. All other temperatures, weights, candidate sets, and supervised rows are derived from the teacher graph. The graph and teacher embeddings are cached before training and are not needed at inference.

### 3.1 Teacher graph

Let $\mathcal{D} = \{x_i\}_{i=1}^{N}$ be the deduplicated training corpus. The teacher encodes every text once, and its embeddings are $\ell_2$-normalized so that all similarities are cosines. For each node $i$, we retrieve the $k = \texttt{graph\_k}$ most similar non-self nodes. If $s_i^{(q)}$ is the $q$-th largest retrieved cosine, the row bandwidth is

$$\tau_i = \frac{s_i^{(1)} - s_i^{(k)}}{\log k}. \tag{1}$$

Thus the $k$-th retrieved neighbor is $k$ times less likely than the nearest neighbor before mutual filtering. Because the bandwidth scales with any affine rescaling of the similarities, the resulting softmax row is affine-invariant and requires no teacher-specific temperature tuning.

We retain an edge $i \to j$ only when $i$ and $j$ retrieve each other. If mutual filtering isolates a node, its raw top-$k$ row is used as a fallback. Writing the resulting neighbor set as $\mathcal{N}_i$, the teacher transition row is

$$P_i^T(j) = \frac{\mathbf{1}[j \in \mathcal{N}_i] \exp(c_{ij}^T/\tau_i)}{\sum_{u \in \mathcal{N}_i} \exp(c_{iu}^T/\tau_i)}. \tag{2}$$

The bandwidth in Equation (1) is computed from the raw top-$k$ scores before this filter, so it is defined from the same number of neighbors for every node.

For numerical efficiency, each row retains the smallest highest-mass prefix $\Omega_i \subseteq \mathcal{N}_i$ satisfying

$$\sum_{j \in \Omega_i} P_i^T(j) \geq 0.99, \qquad \widetilde{P}_i^T(j) = \frac{P_i^T(j)}{\sum_{u \in \Omega_i} P_i^T(u)}. \tag{3}$$

The one-percent tolerance is a numerical-fidelity constant rather than a method hyperparameter. It produces ragged rows while bounding the discarded teacher mass.

### 3.2 DETERMINISTIC CANDIDATE POOL

The candidate set for anchor $i$ is exactly its truncated transition support $\Omega_i$. GGPKD draws no hard or random negatives and performs no per-epoch sampling: the candidates are a deterministic function of the graph and remain fixed throughout training. Within a mini-batch $B$, candidates are deduplicated into the shared pool

$$\mathcal{P}_B = \bigcup_{i \in B} \Omega_i, \tag{4}$$

so each distinct candidate text is encoded once. Ragged rows are padded with their anchor index during collation; a self-mask removes that padding from all softmaxes.

### 3.3 TRAINING OBJECTIVE

The graph-scale loss matches each truncated teacher row with a student softmax over the same columns and at the same stored bandwidth:

$$\mathcal{L}_{r=1} = \frac{1}{|B|} \sum_{i \in B} D_{\mathrm{KL}}\Big(\widetilde{P}_i^T \,\Big\|\, \mathrm{softmax}_{j \in \Omega_i}(c_{ij}^S/\tau_i)\Big). \tag{5}$$

This local row ranks candidates within an anchor's own neighborhood. To also calibrate similarities across the batch, the ambient loss uses the complete shared pool. Its temperature is derived from the graph as $\tau_0 = \mathrm{median}_i\, \tau_i$ and is shared by teacher and student:

$$\mathcal{L}_{r=0} = \frac{1}{|B|} \sum_{i \in B} D_{\mathrm{KL}}\big(\mathrm{softmax}_{j \in \mathcal{P}_B \setminus \{i\}}(c_{ij}^T/\tau_0) \,\big\|\, \mathrm{softmax}_{j \in \mathcal{P}_B \setminus \{i\}}(c_{ij}^S/\tau_0)\big). \tag{6}$$

The ambient and graph groups receive equal total weight; this balance is fixed, not tuned.

GGPKD also reuses non-anchor candidates as auxiliary row centers. Let $\mathcal{J}_B = \mathcal{P}_B \setminus B$ and expose the portion of row $j$ that is already present in the shared pool:

$$\Omega_j^B = \mathcal{N}_j \cap \mathcal{P}_B, \qquad \mathcal{J}_B^+ = \{j \in \mathcal{J}_B : |\Omega_j^B| \geq 2\}. \tag{7}$$

Batch anchors are excluded because $\mathcal{L}_{r=1}$ already supervises their transition rows. The remaining rows are matched uniformly at their stored bandwidths:

$$\mathcal{L}_{\mathrm{row}} = \frac{1}{|\mathcal{J}_B^+|} \sum_{j \in \mathcal{J}_B^+} D_{\mathrm{KL}}\Big(P_j^T|_{\Omega_j^B} \,\Big\|\, \mathrm{softmax}_{u \in \Omega_j^B}(c_{ju}^S/\tau_j)\Big), \tag{8}$$

where the restricted teacher row is renormalized on $\Omega_j^B$. This term requires no additional student encoding because all centers and columns are already in $\mathcal{P}_B$.

The complete objective is

$$\mathcal{L}_{\mathrm{GGPKD}} = \underbrace{\mathcal{L}_{r=0} + \mathcal{L}_{r=1}}_{\mathcal{L}_{\mathrm{rel}}} + \lambda_{\mathrm{row}} \mathcal{L}_{\mathrm{row}}. \tag{9}$$

The defaults are `graph_k` $= 200$ and $\lambda_{\mathrm{row}} =$ `row_weight` $= 1$. Increasing `graph_k` changes both neighborhood width and the bandwidth in Equation (1); those effects cannot be separated by a single-axis sweep.

**Cost.** Exact graph construction requires $O(N^2 d)$ similarity work, although an approximate nearest-neighbor backend can replace the all-pairs search. During training, the dominant cost is encoding the deduplicated shared pool; $\mathcal{L}_{\mathrm{row}}$ reuses those embeddings. At deployment, inference uses only the student encoder.

## 4 EXPERIMENTS

Table 1: Results across three teacher → student settings on nine datasets. Emotion, WiC, and STS-B (marked with ⋆) are in-domain; the remaining datasets are out-of-domain. Bold and underlined values denote the best and second-best results within each setting, excluding the Teacher and Student base reference rows. TALAS and GGPKD results are reported as mean ± standard deviation; the remaining baseline results are from TALAS (Dao et al., 2026). For setting (c), the epoch checkpoint is selected on the validation average.

| | **Classification (F1)** | | | **Pair Classification (AP)** | | | **STS (Spearman)** | | | **Domain Avg** | | **Avg.** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Banking77 | Tweet | Emotion⋆ | MRPC | SciTail | WiC⋆ | SICK | STS12 | STS-B⋆ | Avg-In | Avg-Out | Avg. |
| *(a) Qwen3-Embedding-4B → BERT-base (109M)* | | | | | | | | | | | | |
| Teacher | 93.63 | 72.22 | 68.94 | 84.79 | 91.77 | 66.80 | 83.43 | 82.55 | 86.87 | 74.20 | 84.73 | 81.22 |
| Student base | 84.86 | 47.79 | 68.02 | 77.36 | 67.74 | 59.22 | 42.43 | 21.54 | 20.30 | 42.44 | 60.33 | 54.36 |
| SimCSE-unsup | 89.21 | 69.13 | 53.55 | 82.86 | 76.59 | **71.64** | 68.19 | 61.68 | 70.36 | 65.18 | 74.61 | 71.47 |
| CDM | 89.75 | 70.58 | 53.11 | 84.83 | 75.19 | 71.06 | 69.37 | 68.48 | 73.92 | 66.03 | 76.37 | 72.92 |
| DSKD | 89.66 | 69.93 | 54.51 | 83.13 | 77.95 | 71.53 | 68.65 | 63.37 | 71.42 | 65.82 | 75.45 | 72.24 |
| Jasper and Stella | 89.71 | 73.83 | 65.98 | 85.33 | 80.99 | 70.50 | 73.96 | 67.90 | 75.65 | 70.71 | 78.62 | 75.98 |
| DistillCSE | 90.94 | 70.90 | 60.80 | 82.86 | 78.98 | 68.63 | 75.12 | 72.61 | 77.08 | 68.84 | 78.57 | 75.32 |
| EMO | 90.78 | 73.54 | 67.48 | 84.46 | 81.57 | 69.76 | 75.66 | 68.83 | 77.17 | 71.47 | 79.14 | 76.58 |
| TALAS | $90.44 \pm 0.13$ | $\mathbf{74.62 \pm 0.17}$ | $\mathbf{70.87 \pm 0.52}$ | $\mathbf{85.90 \pm 0.10}$ | $81.78 \pm 0.29$ | $68.84 \pm 0.15$ | $\mathbf{77.56 \pm 0.10}$ | $73.12 \pm 0.30$ | $79.24 \pm 0.10$ | $\mathbf{72.98 \pm 0.19}$ | $80.57 \pm 0.01$ | $78.04 \pm 0.07$ |
| **GGPKD** | $\mathbf{91.98 \pm 0.09}$ | $73.74 \pm 0.26$ | $68.78 \pm 0.30$ | $84.22 \pm 0.13$ | $\mathbf{85.30 \pm 0.16}$ | $66.28 \pm 0.03$ | $77.50 \pm 0.06$ | $\mathbf{75.95 \pm 0.16}$ | $\mathbf{82.15 \pm 0.07}$ | $72.40 \pm 0.10$ | $\mathbf{81.45 \pm 0.12}$ | $\mathbf{78.43 \pm 0.10}$ |
| *(b) BGE-M3 → MiniLMv2-H768 (66M)* | | | | | | | | | | | | |
| Teacher | 93.52 | 73.85 | 68.56 | 85.81 | 91.87 | 61.47 | 79.18 | 78.73 | 84.87 | 71.63 | 83.83 | 79.76 |
| Student base | 81.84 | 47.70 | 71.67 | 79.67 | 65.06 | 60.89 | 49.46 | 26.76 | 24.12 | 44.24 | 62.41 | 56.35 |
| SimCSE-unsup | 87.89 | 71.45 | 55.65 | 83.87 | 76.32 | 67.96 | 66.93 | 59.66 | 63.97 | 62.53 | 74.35 | 70.41 |
| CDM | 89.15 | 70.48 | 58.71 | 84.16 | 75.91 | **70.00** | 68.14 | 62.74 | 67.78 | 65.50 | 75.10 | 71.90 |
| DSKD | 89.51 | 70.53 | 57.27 | 84.39 | 75.36 | 69.96 | 68.81 | 66.78 | 70.05 | 65.76 | 75.90 | 72.52 |
| Jasper and Stella | 88.38 | 74.13 | 62.81 | 85.98 | 80.33 | 68.38 | 74.35 | 66.80 | 74.55 | 68.58 | 78.33 | 75.08 |
| DistillCSE | 90.50 | 72.70 | 59.88 | 84.39 | 78.48 | 69.49 | 73.97 | 70.00 | 73.73 | 67.70 | 78.34 | 74.79 |
| EMO | 90.19 | 74.26 | 61.48 | 85.31 | 80.82 | 68.30 | 75.41 | 67.06 | 76.35 | 68.71 | 78.84 | 75.46 |
| TALAS | $89.72 \pm 0.19$ | $\mathbf{75.38 \pm 0.11}$ | $65.83 \pm 0.58$ | $\mathbf{87.58 \pm 0.02}$ | $82.90 \pm 0.12$ | $66.96 \pm 0.07$ | $75.82 \pm 0.11$ | $68.11 \pm 0.20$ | $77.51 \pm 0.18$ | $70.10 \pm 0.13$ | $79.92 \pm 0.05$ | $76.64 \pm 0.04$ |
| **GGPKD** | $\mathbf{90.78 \pm 0.09}$ | $73.57 \pm 0.32$ | $\mathbf{66.45 \pm 0.20}$ | $85.13 \pm 0.06$ | $\mathbf{83.50 \pm 0.20}$ | $62.75 \pm 0.20$ | $\mathbf{78.27 \pm 0.06}$ | $\mathbf{74.94 \pm 0.18}$ | $\mathbf{81.74 \pm 0.07}$ | $\mathbf{70.31 \pm 0.03}$ | $\mathbf{81.03 \pm 0.08}$ | $\mathbf{77.46 \pm 0.05}$ |
| *(c) Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M)* | | | | | | | | | | | | |
| Teacher | 93.21 | 71.33 | 66.53 | 83.37 | 90.00 | 66.85 | 80.64 | 77.11 | 84.60 | 72.66 | 82.61 | 79.29 |
| Student base | 68.29 | 41.07 | 69.90 | 78.39 | 64.56 | 59.58 | 47.60 | 21.95 | 22.08 | 40.91 | 58.45 | 52.60 |
| SimCSE-unsup | 86.67 | 69.56 | 54.00 | 82.29 | 73.76 | 65.71 | 66.04 | 59.11 | 62.08 | 60.60 | 72.91 | 68.80 |
| CDM | 86.07 | 70.32 | 53.04 | 81.97 | 72.97 | 68.34 | 65.97 | 61.12 | 65.75 | 62.38 | 73.07 | 69.51 |
| DSKD | 85.66 | 69.86 | 52.25 | 82.17 | 73.52 | 68.58 | 66.54 | 62.52 | 66.70 | 62.51 | 73.38 | 69.76 |
| Jasper and Stella | 85.94 | 71.25 | 57.99 | 83.36 | 76.17 | 68.25 | 70.57 | 63.81 | 70.11 | 65.45 | 75.18 | 71.94 |
| DistillCSE | 88.03 | 69.87 | 54.24 | 83.20 | 77.66 | 67.88 | 70.09 | 68.19 | 71.53 | 64.55 | 76.17 | 72.30 |
| EMO | 87.03 | 71.67 | 59.63 | 83.45 | 78.17 | **68.89** | 70.81 | 65.78 | 71.42 | 66.65 | 76.15 | 72.98 |
| TALAS | $84.16 \pm 0.15$ | $\mathbf{72.82 \pm 0.11}$ | $61.13 \pm 0.52$ | $\mathbf{84.85 \pm 0.10}$ | $80.91 \pm 0.27$ | $66.32 \pm 0.07$ | $70.72 \pm 0.15$ | $67.80 \pm 0.08$ | $72.39 \pm 0.31$ | $66.61 \pm 0.24$ | $76.88 \pm 0.07$ | $73.46 \pm 0.10$ |
| **GGPKD** | $\mathbf{88.18 \pm 0.03}$ | $70.78 \pm 0.17$ | $\mathbf{62.69 \pm 0.14}$ | $83.71 \pm 0.07$ | $\mathbf{82.77 \pm 0.11}$ | $65.39 \pm 0.01$ | $\mathbf{73.90 \pm 0.08}$ | $\mathbf{74.57 \pm 0.17}$ | $\mathbf{78.22 \pm 0.02}$ | $\mathbf{68.77 \pm 0.04}$ | $\mathbf{78.99 \pm 0.05}$ | $\mathbf{75.58 \pm 0.03}$ |

### 4.1 Sensitivity analysis

We evaluate GGPKD sensitivity on the Qwen3-Embedding-0.6B→MiniLMv2-H384 setting using seeds 42, 43, and 44. Tables 2 and 3 report downstream quality together with the measured resource cost. Increasing `graph_k` from 50 to 400 changes Avg. by only 0.34 points, but the largest setting costs substantially more time and GPU memory. Relative to the default $\texttt{graph\_k} = 200$, the 400-neighbor setting gains 0.05 points while increasing mean training time by 36.8% and peak GPU memory by 54.2%. The row-loss sweep is also flat: Avg. spans 0.18 points, and the small timing differences are not sufficient to establish a causal cost effect. We therefore retain $\texttt{graph\_k} = 200$ and $\lambda_{\text{row}} = 1$ as defaults rather than selecting a new setting on the test results.

## 5 Conclusion

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Table 2: Sensitivity to the teacher-graph neighborhood size for Qwen3-Embedding-0.6B→MiniLMv2-H384. Downstream scores and training wall time are mean ± sample standard deviation over seeds 42, 43, and 44. Training-memory columns report the maximum per-run peak across those seeds; graph-build measurements contain one build per setting. Host and GPU memory are reported in GiB.

| | **Downstream quality** | | | | **Training cost** | | | **Graph-build cost** | | |
|---|---|---|---|---|---|---|---|---|---|---|
| graph_k | Avg-In | Avg-Out | Avg.↑ | Δ Avg. | Wall (s) | Host | GPU | Wall (s) | Host | GPU |
| 50 | $68.29 \pm 0.09$ | $78.82 \pm 0.03$ | $75.31 \pm 0.04$ | $-0.29$ | $231.53 \pm 16.35$ | 9.82 | 7.98 | 36.8 | 2.19 | 6.44 |
| 100 | $68.46 \pm 0.22$ | $79.02 \pm 0.09$ | $75.50 \pm 0.07$ | $-0.10$ | $257.23 \pm 35.31$ | 9.80 | 9.89 | 20.9 | 1.56 | 0.96 |
| 200 (default) | $68.79 \pm 0.07$ | $79.00 \pm 0.03$ | $75.60 \pm 0.04$ | ref. | $329.90 \pm 8.68$ | 10.15 | 11.50 | 21.8 | 1.63 | 0.96 |
| 400 | $68.89 \pm 0.14$ | $79.02 \pm 0.00$ | $\mathbf{75.64 \pm 0.04}$ | $+0.05$ | $451.30 \pm 18.93$ | 10.59 | 17.74 | 28.6 | 2.11 | 0.98 |

Table 3: Sensitivity to the auxiliary row-relation weight for Qwen3-Embedding-0.6B→MiniLMv2-H384 at the default $\texttt{graph\_k} = 200$. Downstream scores and wall time are mean ± sample standard deviation over seeds 42, 43, and 44. Host and GPU columns report the maximum observed per-run peak in GiB.

| | **Downstream quality** | | | | **Training cost** | | |
|---|---|---|---|---|---|---|---|
| $\lambda_{\text{row}}$ | Avg-In | Avg-Out | Avg.↑ | Δ Avg. | Wall (s) | Host | GPU |
| 0 | $68.63 \pm 0.15$ | $78.86 \pm 0.03$ | $75.45 \pm 0.04$ | $-0.15$ | $326.00 \pm 10.06$ | 10.08 | 11.97 |
| 0.25 | $68.73 \pm 0.19$ | $78.91 \pm 0.06$ | $75.52 \pm 0.07$ | $-0.08$ | $331.53 \pm 10.78$ | 10.15 | 11.50 |
| 0.5 | $68.64 \pm 0.11$ | $78.99 \pm 0.10$ | $75.54 \pm 0.03$ | $-0.06$ | $350.43 \pm 19.74$ | 10.14 | 11.50 |
| 1.0 (default) | $68.79 \pm 0.07$ | $79.00 \pm 0.03$ | $75.60 \pm 0.04$ | ref. | $329.90 \pm 8.68$ | 10.15 | 11.50 |
| 2.0 | $68.91 \pm 0.14$ | $78.99 \pm 0.01$ | $\mathbf{75.63 \pm 0.05}$ | $+0.03$ | $301.23 \pm 33.81$ | 10.17 | 11.50 |

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